

Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 14: Electric Current and Its Effects

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	14 — Electric Current and Its Effects
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Distractors reward shallow rules of thumb — 'more cells always helps', 'a complete-looking loop must light', 'a permanent core makes a better bell', 'a bigger fuse is safer', 'copper out-heats nichrome', 'the field is the same above and below a wire' — and punish anyone who does not trace the branch current actually takes, the make-and-break cycle, or which single effect each device exploits. Do not write on this paper — mark answers on your answer sheet.

1. A loop is drawn so current leaves the battery, reaches a junction, then has two paths that rejoin: path one is a bulb in series with a SECOND bulb; path two is a single identical bulb. All three are good. Tracing where current goes, the brightest bulb is most likely:

- (A) bulb one on path one, because it is reached first by the current
- (B) all three equally, because every bulb in any circuit always glows the same
- (C) the lone bulb on path two, because a single bulb offers an easier path than two bulbs one-after-another, so its branch carries more current
- (D) the second bulb on path one, because the last component always gets the most current

2. A diagram shows a battery, then a wire that forks into branch P and branch Q which rejoin. Branch P holds a closed switch only (a bare wire path). Branch Q holds a bulb. A pupil moves the closed switch in P to OPEN. Before opening, the bulb was dark; after opening it lights. The reason is:

- (A) opening a switch always sends more current everywhere in a circuit
- (B) while P's switch was closed, P was a bare low-resistance shortcut taking nearly all the current; opening P forces the current through the bulb branch
- (C) the bulb needs the switch open in its own branch to glow
- (D) an open switch produces extra push that lights the bulb

3. In a single series loop a pupil places, in order: battery, ammeter-style plain wire, bulb, and a long damp cotton thread, all one-after-another. The bulb glows only very faintly. Removing the damp thread and rejoining the wire makes the bulb bright. This shows the damp thread acted as:

- (A) a branch that drew current away from the bulb
- (B) a source of extra push that should have brightened the bulb
- (C) an open switch that fully broke the loop
- (D) a poor conductor in series, letting only a small current through the whole single loop

4. A circuit symbol shows a battery, then a small gap bridged by a hinged arm resting firmly on both contacts, then a bulb. A pupil labels this 'open switch, bulb off'. The pupil is wrong because:

- (A) a hinged arm symbol always means the switch is open whatever its position
- (B) any gap drawn in a wire keeps the bulb off regardless of the arm
- (C) an arm resting on both contacts is a CLOSED switch, completing the loop, so the bulb glows
- (D) a switch symbol has no effect on whether a bulb lights

5. Two identical bulbs are wired one-after-another (in series) with a battery and a closed switch, both glowing dimly. A pupil joins a thick bare wire directly across the terminals of bulb 1 only. The most likely result is:

- (A) both bulbs go dark, because any extra wire stops all current
- (B) both bulbs brighten equally, because the wire adds push
- (C) bulb 1 goes dark while bulb 2 brightens, because the bare wire bypasses bulb 1 and the loop's current rises
- (D) bulb 2 goes dark while bulb 1 stays the same

6. A pupil reads a circuit drawn as: long-thin line, short-thick line (one cell), then a small circle with a cross inside, then a hinged arm lifted clear of its contact, back to the start. The pupil claims 'the bulb is glowing'. The single fault that makes this wrong is:

- (A) the circle-with-a-cross is a switch, not a bulb
- (B) the lifted hinged arm is an OPEN switch, so the loop is broken and the bulb cannot glow
- (C) the long-thin and short-thick lines mean the cell is dead
- (D) a single cell can never light any bulb

7. A loop has a battery and a bulb in series, plus a separate plain wire joined directly from one battery terminal to the other (a dead short across the whole battery). With everything connected, the most likely outcome is:

- (A) the bulb stays dark and the battery runs down fast, because nearly all the current rushes through the bare short instead of the bulb
- (B) the bulb glows extra-bright, because the short adds a second push
- (C) nothing changes, because an extra wire never affects a circuit
- (D) the bulb glows but the battery is unaffected

8. A single loop holds a battery, a closed switch, and a bulb. A pupil opens the switch for an instant, then closes it again. During the instant the switch is open, the bulb's state and the reason are best described as:

- (A) dark, because an open switch breaks the loop and current stops everywhere at once
- (B) glowing faintly, because the bulb stores some current for a while
- (C) glowing normally, because opening a switch briefly does not affect a bulb
- (D) brighter, because opening the switch builds up push

9. A pupil wires two switches so that BOTH must be closed for the bulb to light (the two switches sit one-after-another in the single loop with the bulb). She closes only one of them. The bulb:

- (A) glows at half brightness, because one closed switch gives half the current
- (B) glows fully, because one closed switch is always enough
- (C) glows brighter, because the open switch saves current for the bulb
- (D) stays dark, because two switches in series both need to be closed to complete the one loop

10. A pupil insists 'any closed-looking ring of wire with a cell and a bulb in it must light the bulb'. A counter-example that proves the pupil wrong is a ring that:

- (A) contains two cells joined the correct way, which would light the bulb
- (B) uses thick copper wire throughout, which would light the bulb
- (C) has the bulb placed before the cell rather than after it, which makes no difference and still lights
- (D) looks complete but contains a strip of dry plastic in series, which blocks the current so the bulb stays dark

11. A bulb glows on two cells joined correctly. A pupil now arranges FOUR identical cells in series but reverses exactly two of them (so two push one way and two the opposite way). Compared with the original two-cell brightness, the bulb will most likely:

- (A) glow far brighter, because four cells always beat two
- (B) stay dark or barely glow, because two pushes forward and two backward cancel, leaving almost no net push

- (C) glow exactly as with the original two cells, because reversing cells never matters
- (D) glow dimmer but still clearly lit, because four cells always leave some push

12. A pupil joins three identical cells in a row in the order forward, reversed, forward, all in series with a bulb. Compared with a SINGLE cell driving the same bulb, the bulb will most likely:

- (A) glow about three times as bright, because three cells are present
- (B) stay completely dark, because one reversed cell cancels everything
- (C) glow about twice as bright, because two of the three face forward
- (D) glow about the same as one cell, because the two forward cells minus the one reversed cell leaves the push of roughly one cell

13. Two cells of different sizes (a big one and a small one) are joined correctly positive-to-negative in series with a bulb, and the bulb glows. A pupil claims the bulb would glow even brighter if the small cell were removed and only the big cell used. The fairest verdict is:

- (A) right, because a big cell alone always beats any pair of cells
- (B) right, because mismatched cells cancel and only the big one works
- (C) wrong; two correctly-joined cells add their pushes, so two cells give more push than the big one alone, and the bulb is brighter with both
- (D) wrong, because removing the small cell would make no difference at all

14. A torch takes two cells. A pupil inserts both cells REVERSED, but both face the SAME way as each other. Pressing the switch, the bulb will:

- (A) glow normally, because two cells still pointing the same way add their pushes and merely send current the other way round the loop
- (B) stay dark, because reversed cells never push
- (C) glow at half brightness, because reversing halves each cell's push
- (D) glow brighter than normal, because reversing doubles the push

15. A pupil builds a battery of two cells and measures a steady glow. She then squeezes a THIRD identical cell in correctly, but it is a nearly-dead cell that pushes only weakly. Compared with the two fresh cells alone, the bulb will most likely:

- (A) glow much brighter, because any third cell adds a full extra push
- (B) go dark, because a weak cell drags the others down to zero
- (C) glow exactly the same, because a weak cell adds nothing whatsoever
- (D) glow a little brighter, because the weak third cell still adds some push the right way, just not a full cell's worth

16. A pupil claims: 'Joining cells side by side touching at their cases (not end to end) makes a battery whose push adds up'. The most accurate verdict is:

- (A) right, because any cells touching anywhere combine their pushes
- (B) right, because the metal cases always carry the push

(C) wrong; cells add their push only when joined in a row, the positive end of one to the negative end of the next, not merely by touching cases

(D) wrong, because cells can never be combined into a battery at all

17. A bulb runs on a two-cell battery. A pupil predicts that simply DOUBLING the number of cells to four (all correct) will always double the brightness, with no limit, each time cells are doubled. The most accurate verdict is:

(A) right, because brightness is always exactly proportional to the number of cells without limit

(B) wrong; adding cells correctly raises the push and usually brightens the bulb, but the filament can only take so much before it burns out, so brightness will not keep doubling forever

(C) wrong, because adding more cells the right way actually dims the bulb

(D) right, because a bulb can survive any number of cells safely

18. A pupil draws a battery symbol inside a loop, but the two cell symbols are drawn facing each other (long-thin line of one touching the long-thin line of the other). With a good bulb and closed switch in the loop, the bulb will most likely:

(A) glow brighter than one cell, because two cell symbols always add up

(B) glow normally, because the way symbols face on paper never affects the circuit

(C) glow at exactly one cell's brightness, because one cell always wins

(D) stay dark or barely glow, because the two cells drawn facing each other push in opposite directions and cancel

19. A nichrome heating coil and a copper connecting wire are in the SAME series loop, so the same current flows through both. A pupil predicts the copper will glow first because copper is the better metal. The prediction is wrong because:

(A) for the same current, the nichrome's higher opposition to current makes it heat up far more, so the nichrome glows while the copper stays cool

(B) copper carries no current, so it cannot heat at all

(C) the current is larger in the nichrome part of the same series loop

(D) copper actually heats more but gives off its heat invisibly

20. Two heating coils are cut from the SAME nichrome wire. Coil X is long and thin; coil Y is short and thick. Each is connected in turn to the same battery. Which glows hotter, and why?

(A) Y, because a short thick wire offers less opposition, so it draws a larger current and produces more heat each second

(B) X, because a longer wire always makes more heat no matter the current

(C) both equally, because they are the same material

(D) X, because thin wires carry more current than thick ones

- 21.** An immersion heater warms a bucket of water. A pupil runs it for 2 minutes instead of 1 minute, with everything else the same. Comparing the water afterwards, the longer run:
- (A) delivers the same total heat, because time does not change heating
 - (B) delivers less heat, because the wire tires and makes less heat over time
 - (C) delivers more total heat to the water, because heat keeps being produced every second the current flows
 - (D) delivers more heat only if the current is also increased
- 22.** A bulb filament (thin coiled tungsten) and its thick copper lead wires carry the SAME current, yet only the filament glows white-hot. The deepest reason is that the filament:
- (A) carries a much larger current than the copper leads do
 - (B) is thin and made of a metal that opposes the current strongly, so the same current dumps far more heat into it than into the thick low-opposition copper
 - (C) is the only part of the circuit that current actually passes through
 - (D) is cooled by the copper leads, which is why the copper stays cool
- 23.** A pupil claims a heater coil and a glowing bulb filament both use the heating effect, so they must reach the same temperature. The fairest verdict is:
- (A) right, because the same effect always gives the same temperature
 - (B) wrong, because a heater uses the magnetic effect, not the heating effect
 - (C) right, because both are made of the same metal
 - (D) wrong; both use the heating effect, but the bulb's filament is driven to a far higher temperature so it glows white, while a low-temperature heater coil only glows dull red
- 24.** In a single series loop, a pupil doubles the current through a fixed nichrome coil by adding cells correctly. Compared with before, the heat produced in the coil each second:
- (A) increases, because more current through the same wire produces more heat each second
 - (B) stays the same, because the wire is unchanged
 - (C) decreases, because faster current spends less time in the wire
 - (D) increases only if the wire is also made thinner
- 25.** Three nichrome coils A, B, C are cut to the SAME length from wires of increasing thickness (A thinnest, C thickest) and connected in turn to the same battery. Ranking the heat produced each second from most to least:
- (A) A, B, C — the thinnest always produces the most heat regardless of current
 - (B) all equal — same length and material means same heat
 - (C) B most, then A, then C — the middle thickness always wins
 - (D) C, B, A — the thickest offers least opposition, draws the most current, and so produces the most heat
- 26.** A pupil argues that a copper connecting wire 'produces no heat at all' when current flows. The most accurate correction is:

- (A) correct, copper is special and produces exactly zero heat
- (B) even copper warms a little, because all ordinary wires produce some heat when current flows; copper simply produces far less than nichrome
- (C) wrong, because copper actually produces more heat than nichrome
- (D) correct, because only coiled wires can produce heat

27. A pupil wants to make a heater coil glow hotter but must keep the SAME single cell. The single most effective change among these is to:

- (A) replace the coil with a much longer, thinner piece of the same nichrome
- (B) coat the coil in plastic so it keeps its heat in
- (C) move the cell further from the coil along the same wire
- (D) replace the coil with a shorter, thicker piece of the same nichrome, so it opposes current less and draws a larger current

28. A compass needle held directly ABOVE a straight current-carrying wire deflects one way. The pupil then carries the SAME wire and current but holds the compass directly BELOW it. The needle will:

- (A) deflect exactly the same way, because the current direction is unchanged
- (B) deflect the OTHER way, because the wire's magnetic effect circles around it, so above and below the wire point in opposite directions
- (C) not deflect at all, because being below a wire shields the compass
- (D) spin continuously, because below a wire there is no fixed direction

29. A pupil places one compass to the LEFT of a vertical current-carrying wire and an identical compass to the RIGHT, both level with it. When current flows, the two needles:

- (A) swing in the same direction, because both feel the same wire
- (B) stay still, because two compasses cancel each other
- (C) swing in opposite directions, because the wire's circling magnetic effect points oppositely on its two sides
- (D) both point straight at the wire

30. A pupil sees a compass under a wire deflect by a small angle. She then joins a second cell correctly in series, raising the current, with wire and compass unmoved. The needle now:

- (A) deflects by a smaller angle, because adding cells weakens the field
- (B) deflects by the same angle, because the field does not depend on current
- (C) deflects by a LARGER angle, because a stronger current makes a stronger magnetic effect
- (D) stops deflecting, because too much current cancels the field

31. In 1820 Oersted noticed a compass needle move near a wire. The crucial condition that produced the movement was that:

- (A) the wire was simply present near the compass, even with no current
- (B) the wire was made of iron rather than copper
- (C) current was actually flowing in the wire, since only a flowing current creates the magnetic effect
- (D) the compass was first stroked with a magnet

32. An engineer must confirm current is flowing inside a sealed, uncut cable. Using an effect of current, the most reliable single test is to:

- (A) look for the cable's insulation changing colour
- (B) listen for the current humming through the insulation
- (C) weigh the cable to see if current adds mass
- (D) bring a small compass close and watch whether its needle deflects, since a current's magnetic effect moves a nearby needle

33. A pupil switches OFF the current in a straight wire under which a compass had been deflecting. The needle will:

- (A) swing back toward its original north-south rest position, because the magnetic effect vanishes the instant the current stops
- (B) stay frozen at its deflected angle, because the wire remembers the current
- (C) deflect even further, because switching off releases stored field
- (D) begin spinning freely with no resting direction

34. A pupil claims that a heater's glowing nichrome coil, because it carries current, must ALSO have a magnetic effect around it while running. This claim is:

- (A) wrong, because a wire can show only one effect at a time
- (B) wrong, because heating destroys the magnetic effect
- (C) correct; any wire carrying current has a magnetic effect around it, even a heating coil — though the heater is built to exploit the heating effect, not the magnetic one
- (D) correct, and it means the heater can be used as a strong electromagnet

35. Two straight wires carry current in the SAME direction; one passes directly OVER a compass and the other directly UNDER the same compass. When current flows, the needle:

- (A) feels nothing, because the two wires always cancel
- (B) spins continuously, because two wires confuse it
- (C) is pulled apart equally and tears off its pivot
- (D) is pushed the same way by both wires, because above one wire and below the other (with same current direction) point the same way, so their effects add

36. An electromagnet on a SOFT-IRON core and an identical coil and current on a HARD-STEEL core of the same shape are both magnets while current flows. After switch-off, the key difference is:

- (A) both lose all magnetism instantly (B) both keep their full magnetism forever
- (C) the soft-iron keeps its magnetism while the steel loses it
- (D) the soft-iron core loses almost all its magnetism at once, while the hard-steel core keeps a good deal of magnetism

37. An electromagnet lifts 6 clips with one cell and 30 turns. A pupil rewinds it to 60 turns but uses much THINNER wire that lets far less current flow. The clips lifted may NOT simply double because:

- (A) strength depends on both turns and current; doubling turns while sharply cutting the current can leave the overall strength little changed
- (B) turns have no effect on an electromagnet's strength
- (C) thinner wire always makes a stronger magnet
- (D) only the core decides the strength, so turns and current are irrelevant

38. A pupil keeps the SAME single cell and SAME soft-iron core but reduces the winding from 40 turns to 20 turns. The electromagnet will most likely now lift:

- (A) fewer pins, because fewer turns make a weaker electromagnet
- (B) more pins, because fewer turns concentrate the current
- (C) the same pins, because turns do not affect strength
- (D) no pins at all, because below 40 turns an electromagnet stops working

39. A pupil winds a coil on a soft-iron core but leaves the two wire ends NOT connected to any cell, then holds the coil near iron pins. The pins are:

- (A) strongly attracted, because the iron core is always a magnet
- (B) not attracted, because with no current there is no magnetic effect at all
- (C) weakly attracted, because the coil stores some magnetism
- (D) repelled, because an unconnected coil pushes iron away

40. A pupil reverses the connections to the cell so current flows the OPPOSITE way through the same electromagnet coil that was lifting iron filings. The electromagnet will:

- (A) stop attracting, because reversed current cannot magnetise
- (B) still attract the iron filings, because reversing the current reverses which end is north but does not stop it being a magnet
- (C) repel the iron filings, because reversing the current repels iron
- (D) become a permanent magnet that no longer needs current

41. A scrap-yard electromagnet must pick ONE metal out of a moving mixture and leave the rest. The mixture it can usefully sort by attracting just one part is:

- (A) aluminium, copper and brass pieces, since the magnet picks the heaviest
- (B) iron and steel only, since a magnet cannot tell them apart so sorts nothing
- (C) plastic and wood, since light items are lifted most easily
- (D) iron rods mixed with aluminium sheets, copper pipes and plastic, since only the iron is attracted

42. Two electromagnets use the same wire, same turns, same single cell. Magnet P has a soft-iron core; magnet Q has a wooden core. Which is stronger, and why?

- (A) Q, because wood is lighter so the field passes through more easily
- (B) P, because a soft-iron core greatly boosts the magnetic effect, while wood adds essentially nothing
- (C) equal, because the core material makes no difference
- (D) Q, because wood insulates the coil and strengthens it

43. An electromagnet holds a chain of nails. With one cell it holds 3 nails; with two cells correctly in series it holds 7. A pupil says doubling the cells MORE than doubled the strength, so strength must rise faster than current forever. The fairest comment is:

- (A) the pupil is exactly right and the lift will keep more-than-doubling each time cells are doubled
- (B) the pupil is wrong because adding cells cannot increase the lift at all
- (C) the strength did rise when current rose, but one jump from 3 to 7 cannot prove it keeps rising faster than current without limit; cores saturate and lifting has limits
- (D) the pupil is wrong because more current weakens an electromagnet

44. A device must become a strong magnet the instant a button is pressed and lose essentially ALL its magnetism the instant the button is released, over and over. The best choice is:

- (A) a steel bar magnet glued to the button, because steel switches on and off easily
- (B) a coil on a hard-steel core, because steel gives instant on-off magnetism
- (C) a coil wound on a soft-iron core fed from a cell through the button, because soft iron magnetises with current and demagnetises at once when current stops
- (D) a plain copper coil with no core, because copper switches magnetism fast

45. In an electric bell the soft-iron core is secretly swapped for a permanent steel magnet that always stays magnetic. Tracing the cycle, the bell will most likely:

- (A) ring faster than before, because a stronger core speeds the cycle
- (B) ring normally, because the bell does not depend on losing magnetism
- (C) pull the armature once and then hold it, failing to ring repeatedly, because a never-demagnetising core keeps gripping after the contact breaks
- (D) refuse to attract the armature at all

46. A pupil glues the bell's make-and-break contact CLOSED so it can never break. Pressing the switch, the bell will:

- (A) ring continuously and louder than ever
- (B) not move the armature at all
- (C) ring faster because the contact never wastes time opening
- (D) give a single strike and then stay silent, because without the contact ever breaking the armature cannot spring back to repeat the cycle

47. In a bell the springy strip carrying the armature is replaced with a STIFF bar that cannot flex. With current on, the bell will most likely:

- (A) fail to ring repeatedly, because the stiff armature cannot spring back to remake contact after it is pulled in
- (B) ring louder, because a stiff bar strikes harder each time
- (C) ring as normal, because springiness is not needed
- (D) produce a steady tone with no striking

48. A pupil files the bell's contact screw so the gap is far too WIDE ever to touch, even with the armature fully at rest. Pressing the switch, the bell will:

- (A) do nothing, because with the contact never closing, no current flows and the electromagnet never pulls
- (B) ring continuously, because a wide gap lets the cycle run freely
- (C) ring once, because the first push closes the gap
- (D) buzz faintly, because current jumps the gap

49. A pupil draws a bell cycle as arrows: current flows -> electromagnet pulls armature -> hammer strikes gong -> contact breaks -> current stops -> ELECTROMAGNET KEEPS FULL MAGNETISM -> armature held forever. The single wrong step is:

- (A) 'current flows' — current never starts the cycle
- (B) 'electromagnet keeps full magnetism' — once the current stops the soft-iron loses its magnetism, releasing the armature so it springs back
- (C) 'hammer strikes gong' — the hammer never reaches the gong
- (D) 'contact breaks' — the contact never breaks in a bell

50. A pupil wires the bell's coil and make-and-break contact normally but uses nearly-dead cells. Pressing the switch, the most likely behaviour is that the bell:

- (A) rings louder than ever, because weak cells make a gentler, clearer tone
- (B) rings only weakly or not at all, because the feeble current makes too weak a pull to work the armature properly
- (C) rings at exactly normal strength, because the bell ignores cell strength
- (D) rings faster, because weak cells switch on and off quicker

51. A pupil builds a bell-style coil-and-soft-iron with a working make-and-break contact, but forgets to fit any hammer or gong. With the switch pressed, what happens?

- (A) nothing moves, because without a gong the cycle cannot run
- (B) the armature still vibrates rapidly back and forth as the contact makes and breaks, but there is no ringing sound since nothing strikes a gong
- (C) the armature is pulled in once and stays stuck
- (D) the coil overheats and melts because the missing gong shorts it

52. A pupil claims: 'Because both a fuse and an electric bell contain a wire the current flows through, both protect the circuit by breaking when the current is too high.' This

claim is:

- (A) right, because any device with a wire protects the circuit
- (B) right, because both break the circuit, so both are safety devices
- (C) wrong; a fuse breaks the circuit by melting to protect against excess current, but a bell's contact breaks repeatedly as part of ringing, not for protection
- (D) wrong, because a bell, not a fuse, is the real protective device

53. A house has three circuits with fuses: lighting (safe up to 5 A), kitchen sockets (15 A), and an air-conditioner (20 A). The correct fitting is:

- (A) a single 20 A fuse on all three, because the largest rating is safest
- (B) a fuse rated just above each circuit's safe limit — about 5 A, 15 A and 20 A respectively — so each melts before its own wiring overheats
- (C) a 5 A fuse on all three, because the smallest rating is safest
- (D) a fuse rated far below each limit, because lower always means safer

54. Someone replaces a proper fuse with a thick nail so it 'never blows'. Two appliances then draw current well above the safe limit, yet nothing trips. The hidden danger is that:

- (A) the wiring now carries the dangerous overcurrent unchecked and can overheat and start a fire, since nothing melts to break the circuit
- (B) the nail will quietly absorb the extra current and make the circuit safer
- (C) the appliances will simply switch themselves off when overloaded
- (D) the current will harmlessly drain to earth through the nail

55. An ideal fuse wire, compared with the household wiring around it, should be made so that for the same large current it:

- (A) heats up and melts at a LOWER current than the wiring can safely carry, so it breaks first and saves the wiring
- (B) carries more current than the wiring without ever melting
- (C) stays cooler than the wiring so it never melts first
- (D) has a higher melting point than the wiring

56. A series loop has a battery, a bulb and a fuse rated just above the normal current. A pupil keeps adding cells correctly until the current rises sharply. The most likely sequence is:

- (A) the bulb gets brighter and brighter with no limit and the fuse stays cool
- (B) the fuse melts first and then the current rises afterward
- (C) current rises -> the fuse heats up -> the fuse melts and breaks the loop -> the bulb goes out
- (D) the battery melts before the fuse, leaving the bulb glowing

57. A pupil compares a blown fuse and a tripped MCB after a fault. The clearest practical difference is:

- (A) a fuse can be reset by switching, while an MCB must be replaced
- (B) a blown fuse must be replaced with a new piece of fuse wire, while a tripped MCB can simply be switched back on
- (C) both must be thrown away after one fault
- (D) neither breaks the circuit; both only sound an alarm

58. A homeowner keeps blowing a 5 A lighting fuse, so they fit a 30 A fuse to 'stop the nuisance'. Why is this a poor idea even though the blowing stops?

- (A) the 30 A fuse lets a current far above the lighting wiring's safe limit flow without melting, so the wiring can overheat and catch fire
- (B) the 30 A fuse will blow even more often than the 5 A one
- (C) a 30 A fuse cannot conduct at all, so the lights will not work
- (D) the larger fuse will make the lights glow far too brightly

59. A pupil tests a fuse wire alone across a strong battery and watches it act. The correct order of what happens is:

- (A) the fuse melts -> then heats up -> then current begins
- (B) the fuse glows -> melts -> then current starts flowing
- (C) the current stops -> the fuse heats -> then it melts
- (D) current flows -> the fuse wire heats up -> it glows -> it melts and breaks, stopping the current

60. A pupil claims a short, thick copper wire makes a great fuse because it can carry huge currents without melting. Why is this exactly backwards?

- (A) a fuse must be the part that melts first to break the circuit; a wire that carries huge currents without melting would never protect anything
- (B) copper cannot conduct electricity, so it cannot be a fuse
- (C) a thick wire melts far too easily to be safe
- (D) fuses must carry more current than the wiring, which thick copper does

Solutions — Science (Class 7), Chapter 14: Electric Current and Its Effects — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	B	21	C	31	C	41	D	51	B
2	B	12	D	22	B	32	D	42	B	52	C
3	D	13	C	23	D	33	A	43	C	53	B
4	C	14	A	24	A	34	C	44	C	54	A
5	C	15	D	25	D	35	D	45	C	55	A
6	B	16	C	26	B	36	D	46	D	56	C
7	A	17	B	27	D	37	A	47	A	57	B
8	A	18	D	28	B	38	A	48	A	58	A
9	D	19	A	29	C	39	B	49	B	59	D
10	D	20	A	30	C	40	B	50	B	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Path two has just one bulb; path one forces current through two bulbs in a row, which is a harder path. The branch with the easier path carries more current, so the single bulb on path two is brightest. Order of arrival does not decide brightness, since current is not 'used up' on the way. The two-bulb branch is harder, so its bulbs are dimmer, not brighter at the end. And bulbs in different branches need not glow equally.

2. (B) A closed bare-wire branch beside the bulb is a short-circuit path that steals almost all the current, so the bulb stays dark. Opening that switch removes the shortcut, so the only remaining path is through the bulb, which now lights. Opening a switch does not raise current everywhere; here it merely redirects it. The switch is in branch P, not the bulb's branch. An open switch makes no push of its own.

3. (D) Everything is in ONE series loop, so the same current must pass through every part. A damp thread is a poor conductor, so it limits the current through the whole loop,

dimming the bulb. It is not a branch, since the loop has no fork. It adds no push. It did not fully break the loop, because the bulb still glowed faintly.

4. (C) When the hinged arm touches both contacts it bridges the gap, which is the CLOSED state, so current flows and the bulb glows. The pupil confused the resting-on-contacts position with 'open'. The arm symbol means open only when it is lifted off a contact. A bridged gap is not a break. A switch certainly controls whether the bulb lights.

5. (C) A bare wire across bulb 1 is an easy path beside it, so current skips bulb 1's filament and bulb 1 goes dark. With one less bulb effectively in the loop, the path is easier, so more current flows and bulb 2 brightens. The wire does not stop all current; it redirects it past bulb 1. It adds no push. Bulb 2 is still in the loop, so it does not go dark.

6. (B) The circle-with-a-cross is the bulb symbol, and it is present, but the lifted hinged arm is an open switch that breaks the loop, so no current flows and the bulb stays dark. The circle-with-a-cross is indeed the bulb, so that is not the fault. The long-thin and short-thick lines are simply the cell's two terminals, not a sign of a dead cell. A single cell can light a small bulb.

7. (A) A bare wire straight across the battery is a very easy path, so almost all the current pours through it rather than through the bulb, leaving the bulb dark, while the heavy current drains the battery quickly. A short adds no push. An extra bare path certainly does affect the circuit by stealing current. The battery is far from unaffected; it discharges rapidly.

8. (A) An open switch breaks the only loop, so current stops everywhere immediately and the bulb is dark for that instant. A bulb does not store current to keep glowing. Opening the switch definitely affects the bulb by cutting its current. Opening a switch builds up no push; it simply interrupts the path.

9. (D) Switches one-after-another in a single loop are both breaks in that loop; if either is open the loop is incomplete and no current flows, so the bulb is dark. A switch does not give 'half' current when closed; the open one still breaks the loop entirely. One closed switch is not enough when another in series is open. An open switch saves no current; it stops it.

10. (D) A ring can look unbroken yet still not conduct if it includes an insulator such as dry plastic in series, which stops the current and keeps the bulb dark — disproving the claim. Two correctly-joined cells would light the bulb, so that is no counter-example. All-copper wiring would also light it. The order of bulb and cell around a single loop does not matter, so that also lights the bulb.

11. (B) Two cells pushing forward and two pushing backward in the same row oppose each other and cancel, so the net push is near zero and the bulb is dark or barely lit. The mere count of cells does not help when half oppose the rest. Reversing cells very much matters;

it subtracts their push. Four cells do not guarantee leftover push when they are evenly opposed.

12. (D) Two forward pushes minus one backward push leaves the net push of about one cell, so the bulb glows roughly as it does on a single cell. Three cells present does not mean triple push when one opposes. One reversed cell does not cancel everything; it only subtracts its own share. The net is two-minus-one, i.e. one cell's worth, not two cells' worth.

13. (C) Cells joined the correct way add their pushes regardless of size, so two cells together push harder than the big one by itself, making the bulb brighter with both. A single big cell does not beat the correctly-joined pair. Mismatched cells joined correctly do not cancel; they add. Removing one of two series cells does reduce the total push, so it is not 'no difference'.

14. (A) If both cells are reversed but still agree with each other, their pushes still add to the full two-cell amount; the current simply circulates the opposite way, and the bulb glows normally. Reversed cells still push; they are not dead. Reversing does not halve a cell's push. Nor does it double it. What matters is that the two cells agree, which they do.

15. (D) A weak but correctly-facing cell still contributes some push in the right direction, so the total rises a little above the two fresh cells — a small brightening, not a full cell's worth. It does not add a whole cell's push because it is nearly dead. A correctly-joined weak cell does not drag the push to zero; only a reversed cell subtracts. And it does add a little, so the glow is not exactly the same.

16. (C) A battery is built by joining cells in a row, positive terminal of one to negative terminal of the next; only then do their pushes add. Merely touching cases side by side does not connect them the right way. The cases touching does not chain the terminals correctly. And cells certainly can be combined into a battery when joined end-to-end the proper way.

17. (B) More correctly-joined cells raise the push and generally brighten the bulb, but a real filament can only stand so much before it overheats and burns out, so the doubling cannot continue without limit. Brightness is not endlessly proportional to cell count. Adding cells the right way brightens, not dims. And a bulb cannot survive any number of cells; too many destroy it.

18. (D) Long-line to long-line means the cells face each other, so their pushes oppose and cancel, leaving the bulb dark or barely lit. Two cell symbols add only when joined positive-to-negative in a row, not facing each other. The orientation of the symbols does represent the real terminal arrangement, so it matters. They do not settle at one cell's brightness; opposing equal pushes cancel toward zero.

19. (A) Heating depends on the wire's opposition to current; nichrome opposes much more than copper, so for the same current nichrome gets far hotter and glows while copper barely warms. Copper does carry the current (it is in the loop). In a single series loop the current is the same everywhere, not larger in the nichrome. And copper does not secretly out-heat nichrome; it genuinely heats less.

20. (A) A short, thick wire opposes current less, so on the same battery it draws a larger current, and more current means more heat per second, making Y hotter. Length alone does not decide heat; a long thin wire actually limits current. Same material does not mean same heating, because shape changes the opposition. Thin wires carry LESS current, not more, for the same push.

21. (C) Heat is produced continuously while current flows, so running twice as long delivers roughly twice the total heat to the water. Time does change the total heat collected, since heating accumulates. The wire does not 'tire' and make less heat over time at steady current. And more time alone increases total heat even without changing the current.

22. (B) Being thin and of a high-opposition metal, the filament converts the same current into far more heat than the thick, easy-conducting copper, so only the filament glows. The current is the same in both, not larger in the filament. Current passes through the copper too, since they are in series. The copper's coolness is due to its low opposition, not because it 'cools' the filament.

23. (D) Both rely on the heating effect, but they are designed for different temperatures: the bulb filament is pushed white-hot to give light, whereas a heater coil glows only dull red at a lower temperature. Using the same effect does not force the same temperature. A heater genuinely uses the heating effect, not the magnetic one. And they are not the same metal — filaments are tungsten, heaters typically nichrome.

24. (A) For a given heating wire, raising the current raises the heat produced each second, so doubling the current increases the heating. Keeping the same wire does not keep heat fixed when the current changes. Current does not 'spend less time' and cool the wire; more current means more heating. And the heating rises with current alone, without needing to thin the wire.

25. (D) For the same length on the same battery, the thickest wire opposes least, draws the most current, and so makes the most heat: C then B then A. The thinnest does not automatically make the most heat; it limits the current. Same length and material do not force equal heat when thickness differs. There is no rule that a middle thickness wins; heat simply rises with current.

26. (B) Every ordinary wire, copper included, produces at least a little heat when current flows; copper merely produces far less than nichrome, so it feels cool. Copper is not

magically zero-heat. It produces less than nichrome, not more. And coiling is not required for heating; a straight wire heats too.

27. (D) On the same cell, a shorter, thicker coil opposes current less, draws more current, and so glows hotter. A longer, thinner piece would oppose more and draw less current, glowing cooler. Coating in plastic would not raise the heat produced and would risk melting; it does not increase current. Moving the cell along the same wire changes nothing about the current.

28. (B) The magnetic effect of a straight current wraps around the wire, so the direction above the wire is opposite to the direction below it; the needle therefore swings the other way. It does not swing the same way, because the field reverses across the wire. Being below the wire does not shield the compass. And the needle settles to a steady deflection, not endless spinning.

29. (C) Because the magnetic effect circles the wire, the left side and the right side point in opposite directions, so the two needles deflect oppositely. They do not swing the same way, since the field reverses across the wire. Two separated compasses do not cancel each other; each responds to the local field. And the needles align with the circling field, not straight at the wire.

30. (C) A larger current produces a stronger magnetic effect, so the needle swings through a bigger angle. Adding cells correctly strengthens, not weakens, the current and field. The field very much depends on the current, so the angle is not unchanged. And more current does not cancel the field; it strengthens it.

31. (C) Oersted's key discovery was that a flowing current, not a mere wire, creates a magnetic effect that moves the needle; with the current off, nothing happens. A wire alone, with no current, does not deflect the needle. The metal of the wire is not the crucial factor. And the effect needs no prior stroking of the compass.

32. (D) A flowing current produces a magnetic effect, so a nearby compass needle will deflect — a clean, non-invasive test through intact insulation. Insulation does not reliably change colour with current. Current does not make an audible hum you can count on through insulation. And current does not add measurable weight to the cable.

33. (A) The current's magnetic effect exists only while current flows; switch it off and the effect vanishes, so Earth's field returns the needle toward north-south. The wire does not 'remember' the current and hold the needle. Switching off releases no stored field to deflect it further. And the needle still has Earth's field to settle on, so it does not spin freely.

34. (C) Every current-carrying wire, including a heater coil, has a magnetic effect around it; the heater is simply designed to use the heating effect. A wire is not limited to one effect at a time. Heating does not destroy the accompanying magnetic effect. But the coil's

magnetic effect is weak and not arranged on an iron core, so it is not a strong electromagnet.

35. (D) Since the field circles each wire, 'below the lower wire-position' and 'above the upper' for same-direction currents end up pointing the same way at the compass, so the two effects add and deflect the needle strongly. They do not automatically cancel. The needle settles to a deflection, not endless spinning. And magnetic effects deflect, they do not physically tear the needle off.

36. (D) Soft iron drops nearly all its magnetism the moment the current stops, which is why it is used in electromagnets; hard steel keeps much of its magnetism, behaving like a permanent magnet. So they do not both lose it instantly, nor do both keep it forever. And it is the steel, not the soft iron, that retains magnetism — the last option has it backwards.

37. (A) An electromagnet's strength rises with both the number of turns and the current; if doubling the turns comes at the cost of a much smaller current, the two changes can roughly offset, so the lift need not double. Turns do affect strength, so they are not irrelevant. Thinner wire (less current) does not strengthen the magnet. And both turns and current matter alongside the core, not the core alone.

38. (A) More turns make a stronger electromagnet, so halving the turns weakens it and it lifts fewer pins. Fewer turns do not concentrate current into extra strength. Turns certainly affect strength, so the lift is not unchanged. And there is no magic threshold at 40 turns; 20 turns still gives a working but weaker magnet.

39. (B) An electromagnet needs current to be magnetic; with the ends unconnected no current flows, so there is no magnetic effect and the pins are not attracted. A plain soft-iron core is not a permanent magnet on its own. An idle coil stores no magnetism to attract pins. And an unconnected coil neither attracts nor repels.

40. (B) Reversing the current swaps the magnet's north and south ends but it remains a magnet, so it still attracts the iron filings. Reversed current still magnetises the core. A magnet attracts unmagnetised iron whichever way it is poled, so it does not repel the filings. And it does not become permanent; it still needs current.

41. (D) An electromagnet attracts iron (and steel) but not aluminium, copper, or plastic, so it can pull iron out and leave the rest. Aluminium, copper and brass are all non-magnetic, so none would be lifted. Iron and steel are both magnetic, so a magnet would lift both rather than separate them. And magnets work by being magnetic, not by lightness, so plastic and wood are not lifted.

42. (B) A soft-iron core strongly concentrates the magnetic effect, so P is far stronger; wood gives no such boost. Lightness of wood does not help the magnetic effect. The core material makes a large difference, so they are not equal. And wood insulating the coil does nothing to strengthen the magnet.

43. (C) More current did strengthen the magnet, but a single observation cannot establish an unlimited faster-than-current rise; in reality the core eventually saturates and the lift levels off. So the pupil's 'forever' conclusion is unsupported. Adding cells does increase the lift, so it is not zero effect. And more current strengthens, not weakens, an electromagnet.

44. (C) A soft-iron-cored coil controlled by the button is magnetic only while current flows and loses magnetism at once when it stops — exactly the on/off behaviour wanted. A permanent steel magnet cannot be switched off. A hard-steel core keeps magnetism after the current stops, so it does not switch off cleanly. A coreless copper coil is far too weak to be a 'strong' magnet.

45. (C) The bell rings because the electromagnet loses its pull when the contact breaks, letting the armature spring back to remake contact; a permanently magnetic core keeps gripping, so the armature is held and the bell does not repeat. It does not ring faster; it stops repeating. The repeating action depends on the magnetism dropping, so a permanent core breaks it. And a magnet would still attract the armature, so it does grip — that is the problem.

46. (D) If the contact can never break, the electromagnet keeps pulling, the armature stays held against the gong, and after the first strike there is no repeat — so a single strike then silence. It does not ring continuously; the breaking step is exactly what makes it repeat. The armature is still pulled, so it does move once. And it does not ring faster; it stops repeating altogether.

47. (A) The repeating action needs the armature to spring back after the contact breaks; a stiff, non-flexing bar cannot return to remake contact, so the bell cannot keep ringing. A stiff bar does not give louder repeated strikes; it gives none after the first. Springiness IS needed for the return stroke. And it does not hum a steady tone; the cycle simply fails.

48. (A) If the gap is too wide to close even at rest, the contact never makes, so no current flows, the electromagnet never energises, and nothing happens. A wide permanent gap stops the cycle, it does not free it. The armature at rest cannot reach the contact, so there is no first strike. And ordinary low voltage does not jump such a gap, so there is no buzz.

49. (B) When the contact breaks the current stops, and a soft-iron electromagnet then loses its magnetism, releasing the armature — so 'keeps full magnetism / held forever' is the wrong step. Current does start the cycle, so that step is fine. The hammer does strike the gong, that step is correct. And the contact breaking is the essential step that makes the bell repeat, so it is not wrong.

50. (B) Weak cells push only a feeble current, giving a weak magnetic pull that may be too weak to swing the armature, so the bell rings weakly or not at all. Weak cells do not make a louder bell. The bell's pull depends on the current, so cell strength is not ignored. And weak cells do not speed the cycle; they sap the pull.

51. (B) The make-and-break cycle still drives the armature to vibrate rapidly, but with no hammer or gong there is simply no ringing sound. The cycle does not need the gong to run, so the armature does move. It is not held stuck; the contact still breaks each time. And a missing gong does not short or overheat the coil.

52. (C) A fuse melts to break the circuit and protect against dangerous current; a bell's contact breaks repeatedly only to make the armature vibrate and ring, which is not protection. Containing a wire does not make something a protective device. Breaking the circuit for a different purpose (ringing) is not protection. And it is the fuse, not the bell, that protects — so the last option is backwards.

53. (B) Each fuse should be rated just above its circuit's safe current so it melts before that circuit's wiring overheats — matching the ratings to each circuit. A single 20 A fuse on the 5 A lighting would let dangerous current flow there. A 5 A fuse on the 20 A air-conditioner would blow under normal use. And a fuse far below the normal current would keep nuisance-tripping, so 'lower is always safer' is wrong.

54. (A) A thick nail will not melt at the overcurrent, so the excess current keeps flowing through the wiring, which can overheat and ignite — the very risk a fuse prevents. The nail does not 'absorb' current safely. Appliances do not switch themselves off on overload without a fuse. And the nail does not drain current safely to earth; it just conducts the dangerous current onward.

55. (A) A fuse must be the weak link: it should melt at a lower current than the wiring can stand, breaking first and protecting the rest. A fuse that carries more without melting would never protect anything. If it stayed cooler than the wiring it would not melt first. And a higher melting point would make it melt LAST, not first — the opposite of what a fuse needs.

56. (C) As the current climbs past the fuse's rating, the fuse heats, melts, and breaks the loop, so the bulb goes out. The bulb does not brighten without limit; the fuse cuts the circuit. The current must rise first to make the fuse melt, so 'melts then rises' has the order wrong. And it is the fuse, designed as the weak link, that gives way — not the battery.

57. (B) A fuse is destroyed when it blows and must be replaced, whereas an MCB merely trips and can be switched back on after the fault is cleared. The roles are not swapped: the fuse, not the MCB, is the one that is replaced. An MCB is reusable, so not both are thrown away. And both genuinely break the circuit; they do not just sound an alarm.

58. (A) A 30 A fuse will not melt until the current is dangerously high — far above what the lighting wiring can safely carry — so the wiring can overheat and ignite while the fuse sits intact. The bigger fuse blows LESS often, not more, which is exactly why it is unsafe. It does conduct fine. And the fuse rating does not change how brightly the lights glow.

59. (D) Current must flow first; it heats the thin fuse wire, which glows and then melts, breaking the path and stopping the current. Melting cannot precede heating or the start of current. Glowing and melting also cannot happen before any current flows. And the current does not stop before the heating; the heating is what eventually stops it by melting the wire.

60. (A) A fuse works by being the weak link that melts first; a thick copper wire that resists melting would let dangerous current keep flowing and never break the circuit, defeating the purpose. Copper does conduct, so that reason is false. A thick wire melts LESS easily, not too easily. And a fuse must carry LESS than the wiring safely, not more — the last option states the wrong design rule.